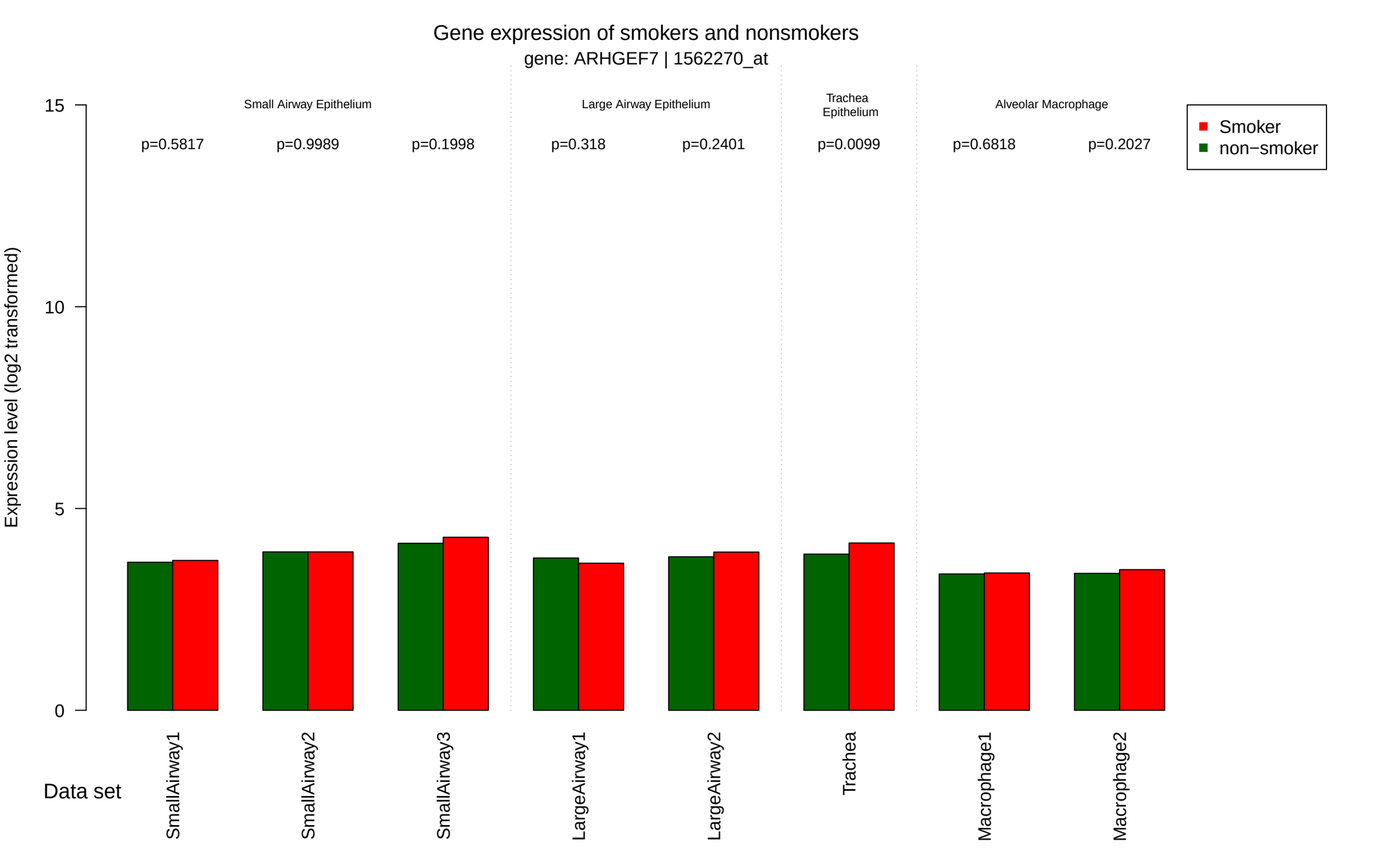
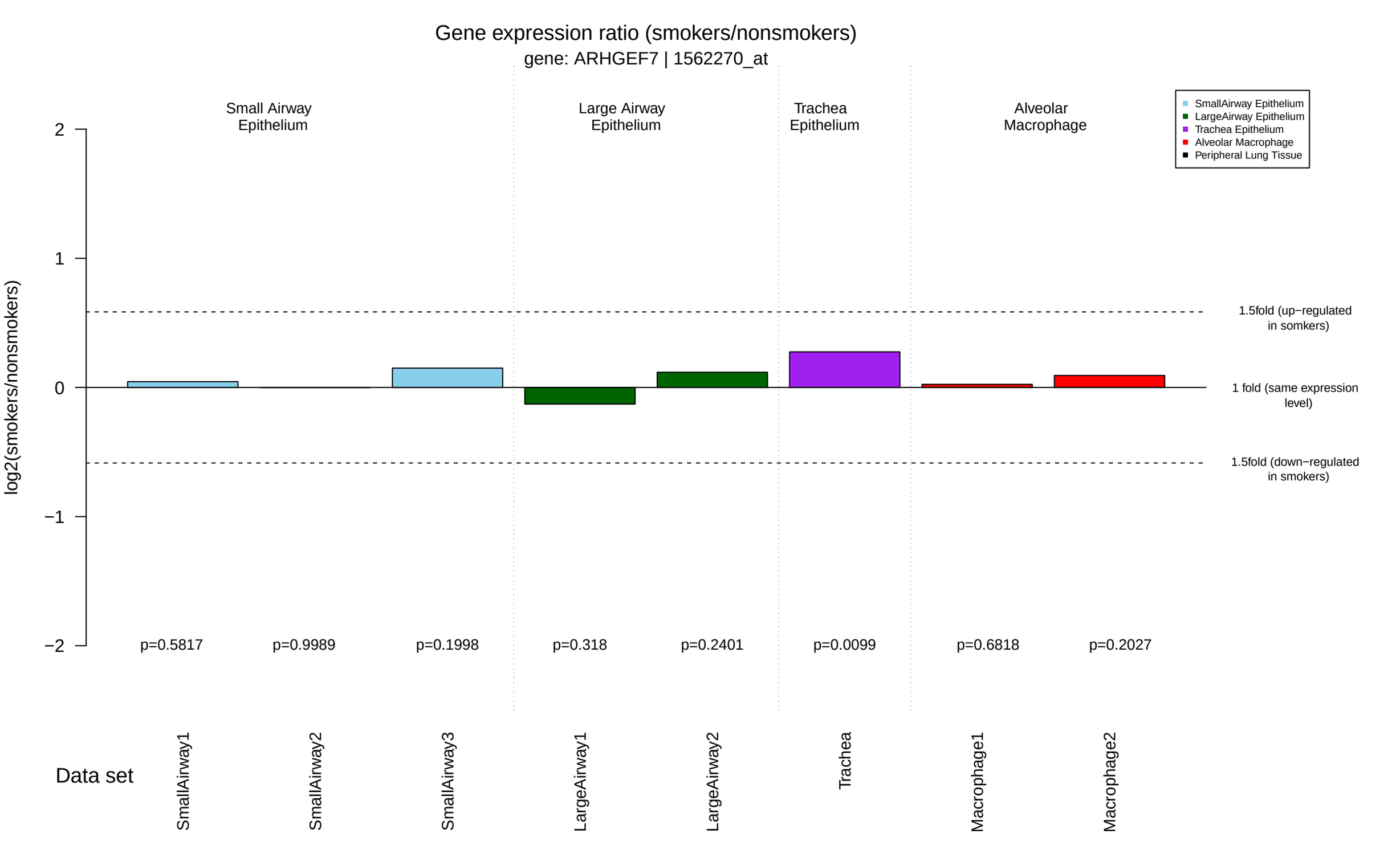
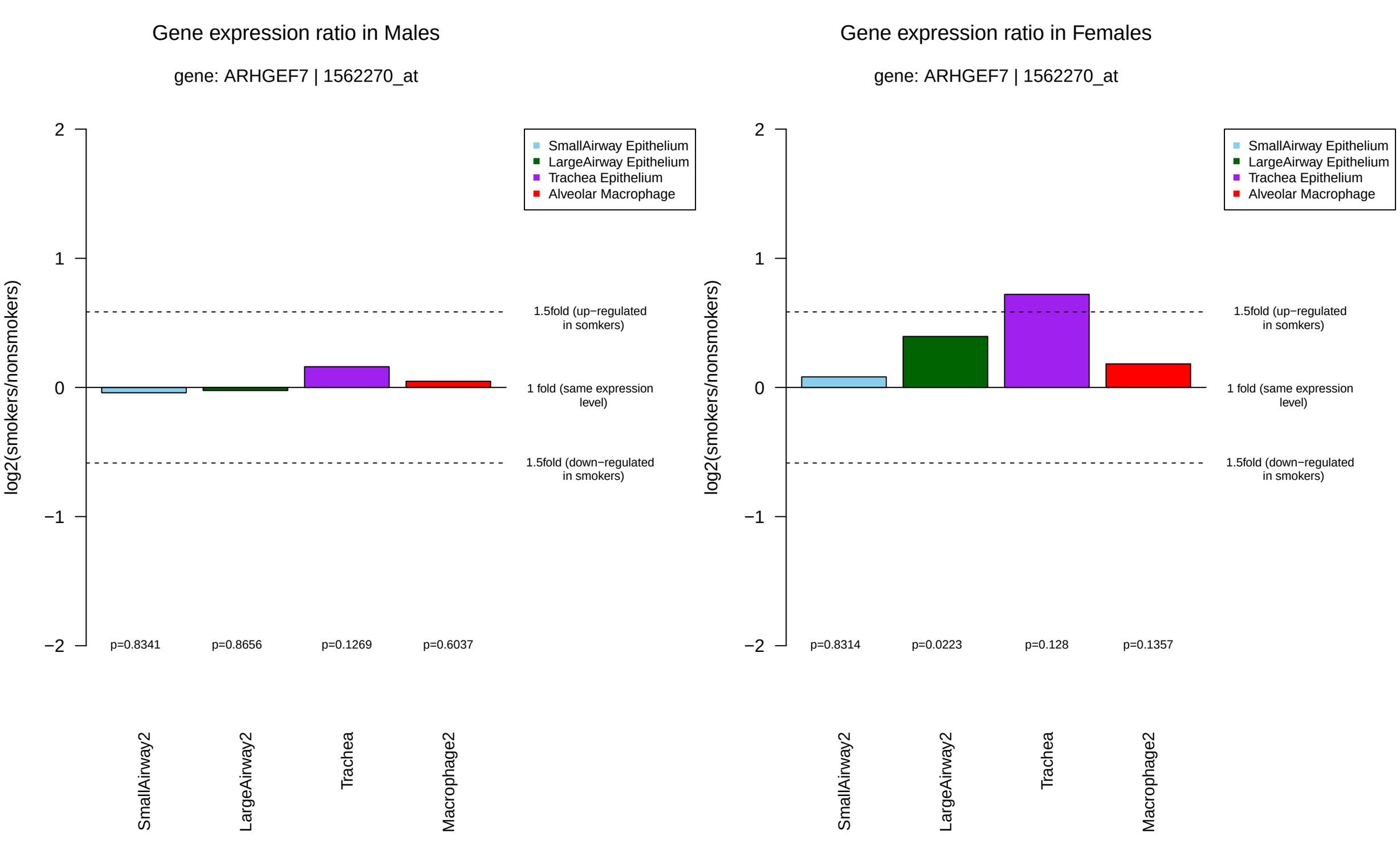
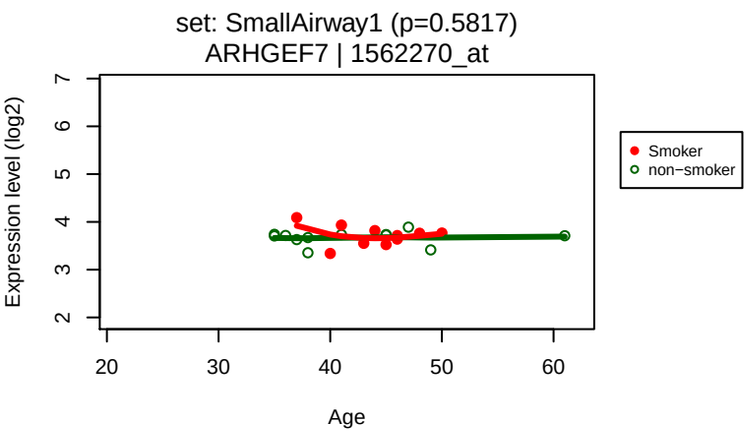
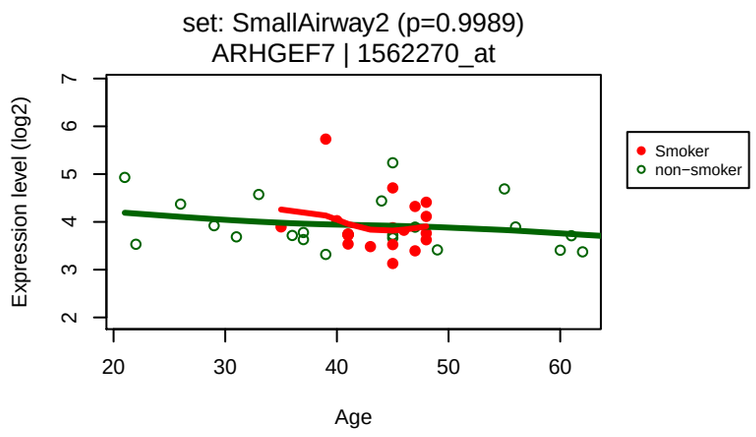
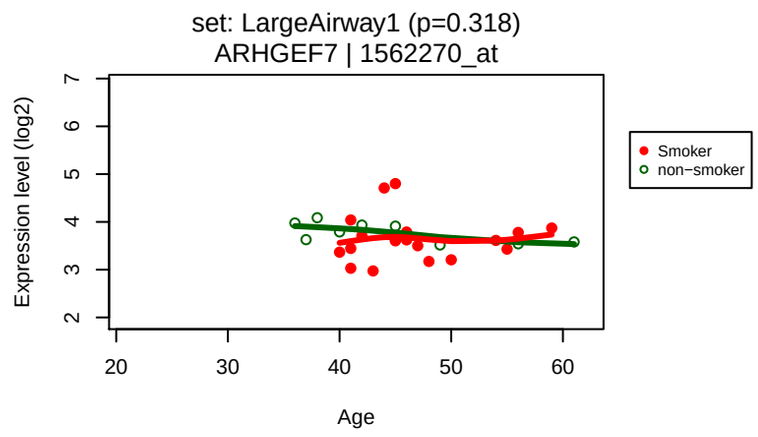

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 1562270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5817
p=0.9989
p=0.1998
p=0.318
p=0.2401
p=0.0099
p=0.6818
p=0.2027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

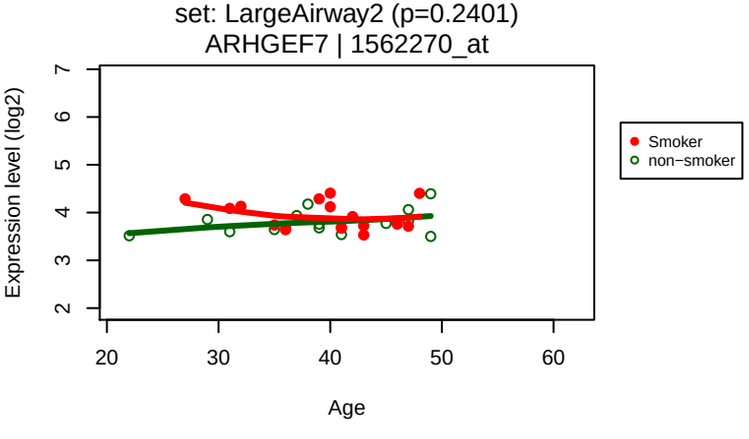

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 1562270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5817
p=0.9989
p=0.1998
p=0.318
p=0.2401
p=0.0099
p=0.6818
p=0.2027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

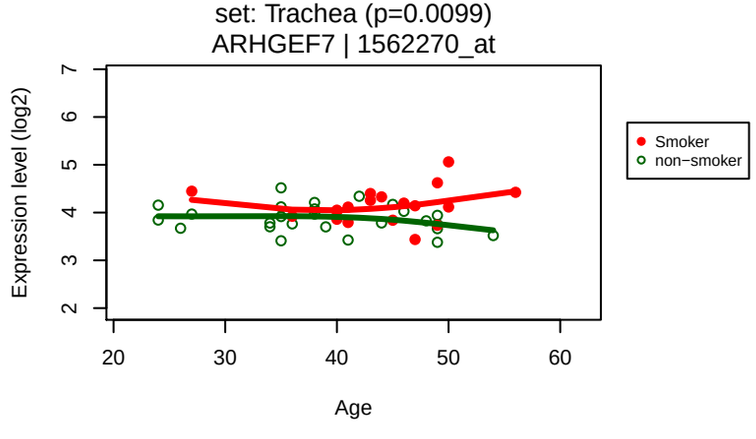

Gene expression ratio in Males
gene: ARHGEF7 | 1562270_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8341
p=0.8656
p=0.1269
p=0.6037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 1562270_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8314
p=0.0223
p=0.128
p=0.1357
SmallAirway2
LargeAirway2
Trachea
Macrophage2

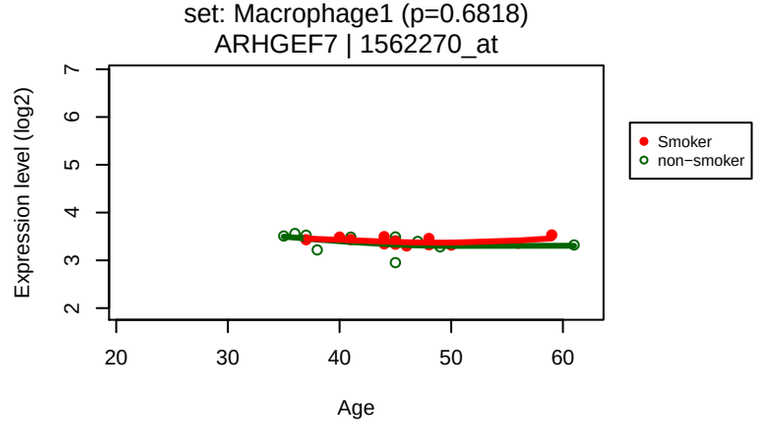

set: SmallAirway1 (p=0.5817)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker

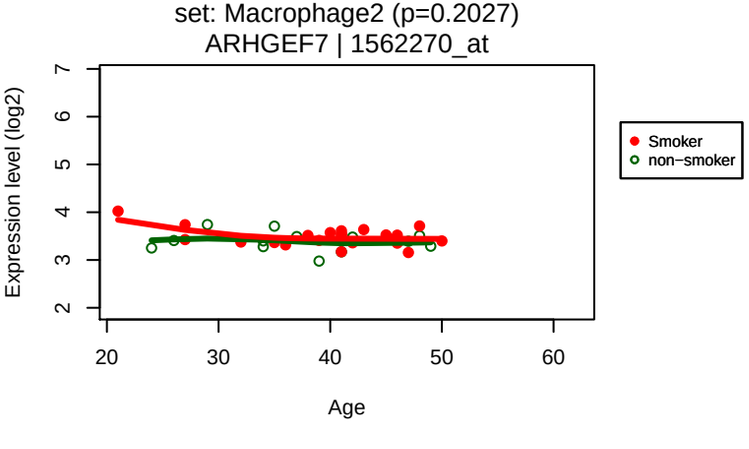

set: SmallAirway2 (p=0.9989)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker

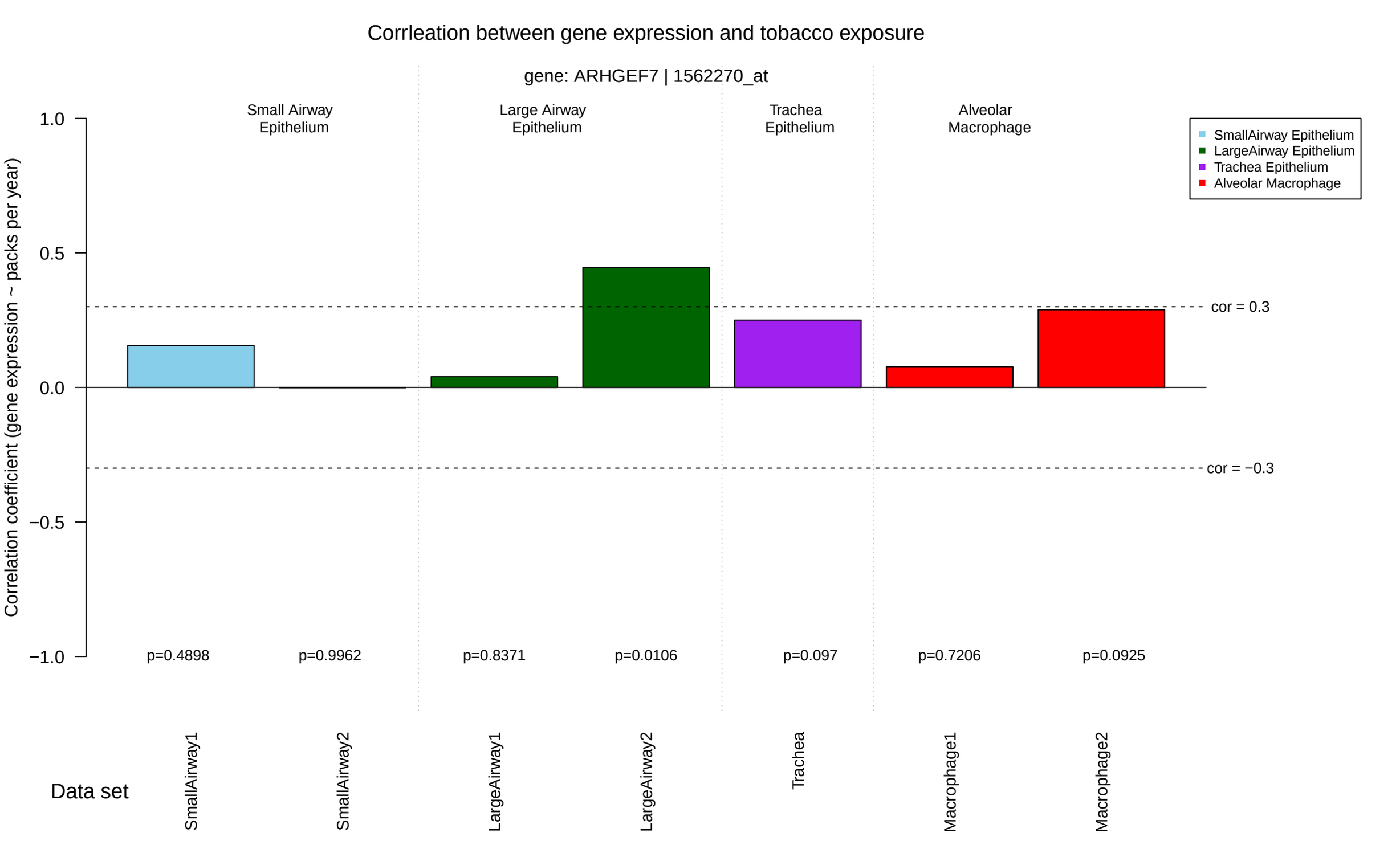

set: LargeAirway1 (p=0.318)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2401)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0099)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6818)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2027)
ARHGEF7 | 1562270_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 1562270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4898
p=0.9962
p=0.8371
p=0.0106
p=0.097
p=0.7206
p=0.0925
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2